VANISHING OF TOR_1 AGAINST THE POWERS OF F FOR A GLUEING PAIR

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For a glueing pair the f-power torsion of R is unchanged under tensoring with R'. invoked at proof lines 25480-25481. Identifies R[f^n] tensor R' with R[f^n].
- The f^n-torsion submodule of R injects into R' for a glueing pair. invoked at proof lines 25481-25482. Gives the injectivity needed for the Tor vanishing.

Source reference. Original lemma 0BNK: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24173–24173. Cited at proof lines 25480-25481 to identify R[f^n] tensor R' with R[f^n]; the conclusion is used inside the counted span.

Source reference. Original lemma 0BNR: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25255–25255. Cited at proof lines 25481-25482 for the injectivity of R[f^n] -> R'; the conclusion is used inside the counted span.

Lemma 0.2. *Let $(R \to R', f)$ be a glueing pair. Then $Tor_1^R(R', f^n R) = 0$ for each $n > 0$.*

Proof. From the exact sequence $0 \to R[f^n] \to R \to f^n R \to 0$ we see that it suffices to check that $R[f^n] \otimes_R R' \to R'$ is injective. By Lemma 0BNK we have $R[f^n] \otimes_R R' = R[f^n]$ and by Lemma 0BNR we see that $R[f^n] \to R'$ is injective as $(R \to R', f)$ is a glueing pair. □

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